PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Update on Noma: Systematic review on classification, outcomes and follow-up of patients undergoing reconstructive surgery after Noma disease
AUTHORS	Speiser, Sophie; Langridge, Benjamin; Birkl, Moira; Kubiena, Harald; Rodgers, Will

VERSION 1 – REVIEW

REVIEWER	Srour, M. Leila Health Frontiers Lao
REVIEW RETURNED	10-Dec-2020

GENERAL COMMENTS	Peer Review: Update on Noma: Systematic review on classification, outcomes and follow-up of patients undergoing reconstructive surgery after Noma disease Reviewed by M. Leila Srour Dear Miss Speiser, Mr. Langridge, Miss Birkl, Mr. Kubiena, and Mr. Rogers, Thank you very much for your work on this important and unique systematic review. I will make some suggestions for your consideration. This article is the first systematic review of the literature on reconstructive surgical outcomes of noma survivors. Mouth opening and complication rates were the two outcomes reported. The analysis of the two noma classification systems was a second objective. The authors have conducted an extensive review of the noma literature, particularly the most neglected area of post-surgical follow-up of noma survivors. Considering how little has been reported and how few patients have had adequate long-term follow-up, I sincerely appreciate their dedication to their objectives. This first review should encourage surgical missions to plan their objectives beyond the surgery and target the local health care workers for post-operative care and follow-up. The suffering of noma survivors is not limited to trismus and surgical complications. Trismus may occur in 50-60% of reported noma survivors and complications of surgery vary on the wide difference of presentations and the experience of the surgical team. The published literature describes how noma's facial deformities leads to stigma, isolation and denial of basic human rights. The Face of Poverty by Klaas Marck describes well the lives

	of noma survivors and the limited possibilities for surgical rehabilitation. The limitation of this review to trismus and complications may significantly miss the positive effects of surgical rehabilitation. This manuscript documents well what is not known, as there are few reports with limited methodological strengths. The lack of follow-up after surgery is the biggest limitation. Significantly, reports of assessment of cosmetic and functional follow-up, especially patients' perceptions are not available. The subjective outcomes section reported a few studies, but without any details. Notably, the three studies listed below showed improvements in the quality of life of noma survivors after surgical rehabilitation. One important point is made (lines 52-56) regarding the role of patient reported outcome measures that should be emphasized. Perhaps rephase the sentence: "If the patient DOES feel that they have benefited"….this may be more relevant than the surgeons' report or the classification system. Underreporting of outcomes and lack of follow-up could lead to UNDER estimation OR over estimation of benefits to the patients. (Lines 23-25) The two classification systems do not appear well designed to compare pre- and post-operative evaluation. As a "control system for disease relapse or progression" (lines 44-47) does not seem appropriate, except for evaluation of trismus. The other sequelae of noma do not relapse or progress, unless surgical complications have negative outcomes. The authors recommend partnerships with local healthcare staff to support post-operative care and long-term follow-up. Validated and standardized questionnaires are essential for pre- and post-op interviews and evaluation for adequate evaluation of surgical outcomes. The authors reported evaluating articles published in the period 1983-2020. However, they did not include two articles published in 2020, documenting follow-up of noma patients after surgery and an important PhD thesis that reports of the quality of life after surgery of noma survivors (unpublished). Perhaps they should specify the exact dates, if they are not including the two published papers by Farley et al and Rickart et al. I will attach the Lafferty thesis, as it has not yet been published and may be difficult to locate. Farley, Elise. "Outcomes at 18 Mo of 37 Noma (Cancrum Oris) Cases Surgically Treated at the Noma Children's Hospital, Sokoto, Nigeria." Transactions of the Royal Society of Tropical Medicine and Hygiene, 2020. https://doi.org/10.1093/trstmh/traa061. Lafferty, N. E. "Changing the Face of Africa: Estimating the Burden of Noma in Rural Ethiopia and Identifying Options for Prevention and Improvement in Its Diagnosis and Management." Liverpool School of Tropical Medicine, 2012. Rickart, Alexander J., Will Rodgers, Kelvin Mizen, Graham Merrick, Paul Wilson, Hiroshi Nishikawa, and David J. Dunaway. "Facing Africa: Describing Noma in Ethiopia." The American Journal of Tropical Medicine and Hygiene 103, no. 2 (August 5, 2020): 613–18. https://doi.org/10.4269/ajtmh.20-0019. Marck, K. W. Noma: The Face of Poverty. Hannover: MIT-Verlag, 2003.

REVIEWER	Bello, Seidu International Craniofacial Academy, Gwarinpa, Abuja, Federal Capital Territory, Nigeria
REVIEW RETURNED	13-Dec-2020

GENERAL COMMENTS	Review Overall Assessment This is a review that is very relevant to the subject of Noma and reconstructive surgery Though my knowledge is not very deep regarding the complex methodology of searching articles for review, but the result is very logical to a surgeon that is very conversant with the topic Am impressed by the comparison of MONTADON and NOITULP classification. Also the vertical and Diagonal approach to Mission activities in Less developed countries. The article is good for publication with minor revision. My Main Observation 1. Generally in the RESULT SECTION, its important to note the following in a scientific write up a. Words must always be used instead of figures at the beginning of every sentence. previously established. 33 articles were excluded, This is not correct. Its beter Thirty three b. 36 Noma patients enrolled per study. 29 studies were case series,……..Twenty-nine is more appropriate c. from 2-140 days, with a mean of 5.5 days. 255 patients were treated Two hundred and fifty-five is more appropriate. d. Below should be considered for better flow and 161 patients (43%) had Montandon Type IV . and 161 (43%) patients had Montandon Type IV . e. Common mistakes: Many authours hardly consider the age of publication while reporting. The earlier should be mentioned before the latter. Eg below under discussion. Montandon et al.(24) based their system on two publications by Cariou(35) in 1986 and Raynaud et al.(36) in 1978. The approach Montandon suggested was rating the defect based upon its severity, taking into account the surgical strategy needed to successfully reconstruct the face. 1978 comes before 1986.

REVIEWER	Adeniyi, Semiu Noma Children Hospital
REVIEW RETURNED	16-Dec-2020

GENERAL COMMENTS	Although the classifications are aids to surgeons but clearly has no benefit for the patient until patient oriented benefit research can be measured all Noma research outcome will be similar

REVIEWER	Prado-Calleros, Héctor Hospital General Dr Manuel Gea González, Education and Research

REVIEW RETURNED	02-May-2021

GENERAL COMMENTS	1. In relation to the inclusion/exclusion criteria possible bias, do the authors were able to identify confounding factors such as: a) if it was a primary surgery or revision surgery, since many of these patients require multiple or staged procedures, an b) if the reconstruction involved only soft tissues or soft tissues and bone reconstruction?; or consider them in the study limitations 2. Did the authors applied other limits in their search strategy, i.e. language? 3. Do the authors have enough data to stratify the outcomes with regards of age of the patients, type of defect (classification) and type of surgery? 4. Could they provide in the discussion section a mention about the fact that most cases of Noma occur between the ages of 2-7 years old, and the median age of the reconstruction was 17 years (range from 4.5-35.7, so the long term evolution may influence the outcome. 5. Thank you

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. M. Leila Srour, Health Frontiers Lao

Comments to the Author:
Peer Review: Update on Noma: Systematic review on classification, outcomes and follow-up of patients undergoing reconstructive surgery after Noma disease

Reviewed by M. Leila Srour

Dear Miss Speiser, Mr. Langridge, Miss Birkl, Mr. Kubiena, and Mr. Rogers,

Thank you very much for your work on this important and unique systematic review. I will make some suggestions for your consideration.

This article is the first systematic review of the literature on reconstructive surgical outcomes of noma survivors. Mouth opening and complication rates were the two outcomes reported. The analysis of the two noma classification systems was a second objective.

The authors have conducted an extensive review of the noma literature, particularly the most neglected area of post-surgical follow-up of noma survivors. Considering how little has been reported and how few patients have had adequate long-term follow-up, I sincerely appreciate their dedication to their objectives. This first review should encourage surgical missions to plan their objectives beyond the surgery and target the local health care workers for post-operative care and follow-up.

The suffering of noma survivors is not limited to trismus and surgical complications. Trismus may occur in 50-60% of reported noma survivors and complications of surgery vary on the wide difference of presentations and the experience of the surgical team. The published literature describes how noma's facial deformities leads to stigma, isolation and denial of basic human rights. The Face of Poverty by Klaas Marck describes well the lives of noma survivors and the limited possibilities for surgical rehabilitation. The limitation of this review to trismus and complications may significantly miss the positive effects of surgical rehabilitation.

Dear Dr. M. Leila Srour, thank you very much for your feedback on the article. In this systematic review, we aimed at including all objective and subjective outcomes after NOMA surgery reported by the published literature we found. We intentionally did not limit our search to any specific outcome measurement methods, mouth opening and complication rates happened to be the most reported outcomes. Surgical rehabilitation is an important aspect of treatment of NOMA disease which most certainly influences the outcome in short- and long-term, but it is neither an outcome itself nor an outcome measurement method of NOMA surgery.
We mentioned that high complication rates could be due to lack of postoperative physiotherapy, which has been reported by Hendrick's et al. before (see Page 11, row 12). We were not able to include The Face of Poverty by Klaas Marck, as it is written in Dutch.

This manuscript documents well what is not known, as there are few reports with limited methodological strengths. The lack of follow-up after surgery is the biggest limitation. Significantly, reports of assessment of cosmetic and functional follow-up, especially patients' perceptions are not available. The subjective outcomes section reported a few studies, but without any details. Notably, the three studies listed below showed improvements in the quality of life of noma survivors after surgical rehabilitation.

Thank you very much for sending us the three studies below, all of them have important insights into NOMA disease, and quality of life is indeed an important outcome after NOMA surgery. We only included articles up to April 15th 2020, unfortunately they are more recent than that. But we will be sure to mention quality of life as an important factor in the discussion.

One important point is made (lines 52-56) regarding the role of patient reported outcome measures that should be emphasized. Perhaps rephase the sentence: "If the patient DOES feel that they have benefited"….this may be more relevant than the surgeons' report or the classification system. Underreporting of outcomes and lack of follow-up could lead to UNDER estimation OR over estimation of benefits to the patients. (Lines 23-25)

Thank you very much for your feedback on this. We have implemented the changes.

The two classification systems do not appear well designed to compare pre- and post-operative evaluation. As a "control system for disease relapse or progression" (lines 44-47) does not seem appropriate, except for evaluation of trismus. The other sequelae of noma do not relapse or progress, unless surgical complications have negative outcomes.

Thank you very much for your feedback on this, we specified now that the classification systems are mainly useful for pre- and postoperative evaluation of trismus patients.

The authors recommend partnerships with local healthcare staff to support post-operative care and long-term follow-up. Validated and standardized questionnaires are essential for pre- and post-op interviews and evaluation for adequate evaluation of surgical outcomes.

Thank you very much for your feedback on this, we added the validated and standardized questionnaires as an essential tool for surgical evaluation.

The authors reported evaluating articles published in the period 1983-2020. However, they did not include two articles published in 2020, documenting follow-up of noma patients after surgery and an important PhD thesis that reports of the quality of life after surgery of noma survivors (unpublished).

Perhaps they should specify the exact dates, if they are not including the two published papers by Farley et al and Rickart et al. I will attach the Lafferty thesis, as it has not yet been published and may be difficult to locate.

We changed the date to specify our search included articles from January 1st 1983- April 15th, 2020. Unfortunately, we cannot include the Laffey thesis, as we only included published articles in our systematic review.

Farley, Elise. “Outcomes at 18 Mo of 37 Noma (Cancrum Oris) Cases Surgically Treated at the Noma Children’s Hospital, Sokoto, Nigeria.” Transactions of the Royal Society of Tropical Medicine and Hygiene, 2020. https://doi.org/10.1093/trstmh/traa061.
Lafferty, N. E. “Changing the Face of Africa: Estimating the Burden of Noma in Rural Ethiopia and Identifying Options for Prevention and Improvement in Its Diagnosis and Management.” Liverpool School of Tropical Medicine, 2012.
Rickart, Alexander J., Will Rodgers, Kelvin Mizen, Graham Merrick, Paul Wilson, Hiroshi Nishikawa, and David J. Dunaway. “Facing Africa: Describing Noma in Ethiopia.” The American Journal of Tropical Medicine and Hygiene 103, no. 2 (August 5, 2020): 613–18. https://doi.org/10.4269/ajtmh.20-0019.
Marck, K. W. Noma: The Face of Poverty. Hannover: MIT-Verlag, 2003.

Thank you very much for your support for survivors of noma and this neglected tropical disease.

Please also see attachments

With warm regards,
M. Leila Srour
MD FAAP MPH DTM&H

Reviewer: 2
Dr. Seidu Bello, International Craniofacial Academy, Gwarinpa, Abuja, Federal Capital Territory, Nigeria

Comments to the Author:
Review
Overall Assessment
This is a review that is very relevant to the subject of Noma and reconstructive surgery

Though my knowledge is not very deep regarding the complex methodology of searching articles for review, but the result is very logical to a surgeon that is very conversant with the topic

Am impressed by the comparison of MONTADON and NOITULP classification. Also the vertical and Diagonal approach to Mission activities in Less developed countries.

The article is good for publication with minor revision.
My Main Observation
1. Generally in the RESULT SECTION, its important to note the following in a scientific write up
a. Words must always be used instead of figures at the beginning of every sentence.
previously established. 33 articles were excluded, This is not correct. Its beter Thirty three

b. 36 Noma patients enrolled per study. 29 studies were case series,……..Twenty-nine is more appropriate

c. from 2-140 days, with a mean of 5.5 days. 255 patients were treated
Two hundred and fifty-five is more appropriate.
d. Below should be considered for better flow
and 161 patients (43%) had Montandon Type IV .
and 161 (43%) patients had Montandon Type IV .

e. Common mistakes: Many authours hardly consider the age of publication while reporting. The earlier should be mentioned before the latter. Eg below under discussion.

Montandon et al.(24) based their system on two publications by Cariou(35) in 1986 and Raynaud et al.(36) in 1978. The approach Montandon suggested was rating the defect based upon its severity, taking into account the surgical strategy needed to successfully reconstruct the face.

1978 comes before 1986.

Best of Luck

Dear Dr. Seldu Bello, thank you very much for your feedback. We have implemented the suggested changes into the revised article version.

Reviewer: 3
Dr. Semiu Adeniyi, Noma Children Hospital

Comments to the Author:
Although the classifications are aids to surgeons but clearly has no benefit for the patient until patient oriented benefit research can be measured all Noma research outcome will be similar

Dear Dr. Semlu Adenlyl, thank you very much for your feedback. We agree that the classification systems are not patient oriented. We mentioned the patient-reported outcome measures and included validated and standardized questionnaires in the revised version.

Reviewer: 4
Dr. Héctor Prado-Calleros, Hospital General Dr Manuel Gea González

Comments to the Author:
1. In relation to the inclusion/exclusion criteria possible bias, do the authors were able to identify confounding factors such as:
a) if it was a primary surgery or revision surgery, since many of these patients require multiple or staged procedures, an

b) if the reconstruction involved only soft tissues or soft tissues and bone reconstruction?; or consider them in the study limitations

2. Did the authors applied other limits in their search strategy, i.e. language?

3. Do the authors have enough data to stratify the outcomes with regards of age of the patients, type of defect (classification) and type of surgery?

4. Could they provide in the discussion section a mention about the fact that most cases of Noma occur between the ages of 2-7 years old, and the median age of the reconstruction was 17 years (range from 4.5-35.7, so the long term evolution may influence the outcome.

Dear Dr. Hector Prado-Calleros, thank you very much for your feedback. We initially considered primary surgery versus revision surgery and local/ regional flaps versus free flaps in our data collection sheet. Unfortunately, there was not enough data published to consider these two confounding factors, or to stratify the outcomes with regards of age of the patients, type of defect and type of surgery. We added the surgical methods as a limitation of this study. We only considered articles that were written in English or German. We added that to the exclusion criteria in the supplementary material. We have also implemented the age range into the discussion.

VERSION 2 – REVIEW

REVIEWER	Srour, M. Leila Health Frontiers Lao
REVIEW RETURNED	03-Jul-2021

GENERAL COMMENTS	Dear Miss Speiser, Mr. Langridge, Miss Birkl, Mr. Kubiena, and Mr. Rogers, Thank you for responding to my comments and the revision of this manuscript. The Face of Poverty book by Klaas Marck was published in English, but the publishing company has closed. You may be able to obtain the book from Dr. Marck at k.marck@chello.nl Normally noma is not capitalized as it is the name of a disease like typhoid and does not require capitalization, unless at the beginning of a sentence or as part of the name of a hospital. Another challenge of doing research for short term missions is the need for ethical review board approval from the host country. Advance planning, recurrent mission trips, and relationships with the host country's academic research programs can be successful. As noma patients are known to be extremely poor and from remote areas, follow-up can be more successful if the patients are told that their travel expenses will be covered. There should be some incentive for them to return, otherwise if they are unhappy or happy with the results, in either case, they may be unlikely to return for follow-up. The diagonal treatment strategy has good potential. Thank you for your research and for the revised manuscript

REVIEWER	Prado-Calleros, Héctor Hospital General Dr Manuel Gea González, Education and Research
REVIEW RETURNED	03-Jul-2021

GENERAL COMMENTS	This systematic review provides a timely and necessary synthesis of reconstructive surgery results for NOMA disease. The revision recommendations have been addressed correctly. Thank you.